

01-ccna-notes

Monday, August 03, 2026 2:23 PM

Notes from Jeremy's IT Lab (Days 1-10)

802.3 Ethernet Standards:

10Mbps	10BASE-T	802.3i
100Mbps	100BASE-T	802.3u
1Gbps	1000BASE-T	802.3ab
10Gbps	10GBASE-T	802.3an

Copper UTP wiring can be used up to 100m

10BASE-T	2 twisted pairs, 4 total wires	(1-2, 3-6)
100BASE-T	2 twisted pairs, 4 total wires	(1-2, 3-6)
1000BASE-T	4 twisted pairs, 8 total wires	(1-2, 3-6, 4-5, 7-8)
10GBASE-T	4 twisted pairs, 8 total wires	(1-2, 3-6, 4-5, 7-8)

Full-Duplex transmission allows devices to transmit (Tx) and receive (Rx) data at the same time.

PC	Tx 1-2, Rx 3-6
Router	Tx 1-2, Rx 3-6
Firewall	Tx 1-2, Rx 3-6
Switch	Tx 3-6, Rx 1-2

Key idea:

Devices that share the same pin layout must use a crossover cable – devices that have an opposite pin layout must use a straight-through cable.

Straight-through cables connect straight through from one pin to another pin

(e.g., 1:1 2:2 3:3 6:6).

Crossover cables reverse the pin mappings to enable communication between devices with the same Tx/Rx pin layout (e.g., 1:3 2:6 3:1 6:2).

Auto MDI-X allows devices to automatically detect which pins their neighbor is transmitting data on and then adjust which pins they use to transmit/receive data with.

- With 1000BASE-T & 10GBASE-T (4 twisted pairs, 8 wires total) each wire is bidirectional, meaning no one pin is dedicated solely to transmitting/receiving data - rather they can function as needed.

802.3 Fiber Standards:

1Gbps	1000BASE-LX	802.3z	MM 550m, SM 5km
10Gbps	10GBASE-SR	802.3ae	MM 400m
10Gbps	10GBASE-LR	802.3ae	SM 10km
10Gbps	10GBASE-ER	802.3ae	SM 30km

SFP transceivers are used to connect fiber-optic connections to network devices.

Multimode fiber has a wider core diameter compared to single mode fiber. This allows multiples angles (modes) of light waves to enter the fiberglass core.

A **protocol** is a set of rules defining how data should be communicated between devices over a network. This can be thought of as the “languages” computers use to communicate.

A **standard** is the agreed upon specification that describes how a protocol or technology should work. With vendor neutral standards, devices of all types can communicate with each other.

TCP/IP Model

Application	Protocols for communication between application processes; create and interpret data (e.g., HTTPS, FTP, SMTP).
Transport	Provides communication between application processes using port numbers (e.g., TCP, UDP).
Network	Provides communication between hosts across networks using IP addresses and routers (e.g., IPv4, IPv6, ICMP).
Data Link	Provides hop-to-hop delivery within a local network using MAC addresses and switches (e.g., 802.3, 802.11).
Physical	Sends bits as electrical, optical, or radio signals over the physical medium.

Encapsulation: As data moves down the TCP/IP layers, each layer adds its own header (layer 2 adds a header/trailer). This is how each layer communicates with its counterpart on the receiving device.

Decapsulation: On the receiving side, each layer removes its corresponding header, Layer 2 also checks/removes the trailer, and processes the information.

PDU (Protocol Data Unit)

Layer 5 (Application)	Data
Layer 4 (Transport)	Segment/Datagram
Layer 3 (Network)	Packet
Layer 2 (Data-link)	Frame
Layer 1 (Physical)	Bit

The contents of each PDU (everything encapsulated by that layer’s header/trailer) is called the **payload**:

A segment/datagram’s (L4) payload is the application data (L5)

A packet’s (L3) payload is a segment/datagram (L4)

A frame’s (L2) payload is a packet (L3)

Adjacent-layer interaction describes how each layer provides a service to the layer above it, and is serviced by the layer below it:

- Layer 4 provides a service to Layer 5 by delivering data to the correct application using port numbers.
- Layer 3 provides a service to Layer 4 by delivering segments/datagrams to the correct destination host using IP addresses.
- Layer 2 provides a service to Layer 3 by delivering packets to the next hop using MAC addresses.
- Layer 1 provides a service to Layer 2 by sending and receiving frames as electrical, optical, or radio signals.

Same-layer interaction describes how each layer communicates with the same layer on other devices:

- The Application layer on one host sends data to the Application layer on the other host.
- A segment/datagram is addressed to the Layer 4 port number of the correct application on the destination host.

- A packet is addressed to the Layer 3 IP address of the destination host.
- A frame is addressed to the Layer 2 MAC address of the next hop.
- Signals sent out of a physical port are received by a physical port on the connected device.

Configuring a new Cisco device:

First, connect through the console port using a rollover cable. Rollover cables usually have a DB9 connector on one end and an RJ45 connector on the other end (you'll likely need a DB9 to USB adapter to connect using your laptop).

Once you're connected through the console port you'll need to use a terminal emulator to access the CLI (PuTTY is a popular choice).

If you need to configure the default CLI settings, Open PuTTY > Click 'Serial'

Cisco CLI default settings:

Speed (baud)	9600
Data bits	8
Stop bits	1
Parity	None
Flow Control	None

Inside the Cisco CLI, there are 3 main modes:

- **User EXEC Mode >**, Basic access mode used to view limited device information and run simple commands. You cannot make configuration changes.
- **Privileged EXEC Mode #**, Administrative mode that gives access to more advanced troubleshooting and management commands. From here, you can enter configuration mode.
- **Global Configuration Mode (config)#**, Used to make configuration changes that affect the device. From here, you can also enter more specific configuration modes such as interface or routing configuration.

When you first enter the Cisco CLI, you will by default be in User EXEC Mode. To enter into Privileged EXEC Mode, you must run:

- enable

To view a list of available command inputs, run:

- ?

(Entering it immediately following a command, without a space, will display all variations of what you've entered. Placing a space between the command and the ? will display everything that can be used in conjunction with the initial command.)

Another useful feature of the Cisco CLI is command completion.

- When entering a command such as enable, you can type en and press Tab to automatically complete the command. You can also press Enter after typing en, and the CLI will recognize it as enable as long as the abbreviation is unique.
- A limitation is that the abbreviation must be specific enough to identify only one command. For example, entering e may return a '% Ambiguous command' error because multiple available commands begin with e.
- To view the commands that begin with a specific letter or abbreviation, type ? directly after it. For example, entering e? displays the available commands that begin with e.

If you'd like to protect Privileged EXEC Mode with a password, from Global Configuration Mode run:

- enable password *password*

Once someone consoles into this device and attempts to enter 'enable' to launch Privileged Exec Mode, they will be prompted for a password - while

typing in the password it will not be displayed for privacy reasons. Inputting the incorrect password 3 times will trigger the error message '% bad secrets'.

There are two configuration files kept on the device at any given time:

- running-config, The current, active configuration file on the device. As you enter commands in the CLI, you edit the active configuration.
- startup-config, The configuration file that will be loaded upon restart of the device.

To view the running configuration file, run:

- show running-config

To view the startup configuration file, run:

- show startup-config

To save the current running configuration file (make it the startup configuration file), run any of these:

- write
- write memory
- copy running-config startup-config

To hide the Privileged Exec Mode password from being displayed in the startup-config files, run:

- service password-encryption

(Important to note that this command can provide encryption, but the encryption itself can be cracked easily.)

To set a password with more advanced encryption (MD5 encryption), and that is automatically hidden from startup-config files, run:

- enable secret *password*

Important:

If we're in Global Configuration Mode and need to run a command that would normally only be available in Privileged EXEC Mode, we can place 'do' in front of the command. This allows the command to be executed without leaving the current configuration mode.

To cancel or delete a command, run:

- no *original-command*

(e.g., no shutdown)

Ethernet Frames

Ethernet frames contain both a header and a trailer.

Header	Preamble, SFD, Destination MAC Address, Source MAC Address, Type/Length
Trailer	Frame Check Sequence (FCS)

Preamble

Length	7 bytes / 56 bits
Function	Allows devices to synchronize their receiver clocks to ensure they're ready to receive the rest of the frame and the data inside.
Format	Alternating 1s and 0s, 10101010 7 times

SFD (Start Frame Delimiter)

Length	1 byte / 8 bits
Function	Marks the end of the Preamble, and the beginning of the Ethernet frame.
Format	10101011

Destination MAC Address

Length	6 bytes / 48 bits
Function	Identifies the intended recipient.
Format	48-bit MAC address

Source MAC Address

Length	6 bytes / 48 bits
Function	Identifies the sender.
Format	48-bit MAC address

Type/Length

Length	2 bytes / 16 bits
Function	The field can indicate either the length of the Ethernet payload or the type of protocol encapsulated in the payload. A value of 1500 or less indicates Length, measured in bytes. A value of 1536 or greater indicates Type (EtherType).
Format	IPv4 = 0x0800 = 2048 IPv6 = 0x86DD = 34525

FCS (Frame Check Sequence)

Length	4 bytes / 32 bits
Function	Detects corrupted data by running a CRC algorithm over the received data.

Important:

The Preamble and SFD are usually not considered part of the Ethernet header.

Therefore, the Ethernet header and trailer together total 18 bytes:

- Header: 14 bytes (6 + 6 + 2)
- Trailer (FCS): 4 bytes
- Total: 18 bytes

The minimum Ethernet frame size is 64 bytes, consisting of the header, payload, and trailer.

64 bytes - 18 bytes = 46 bytes

Therefore, the minimum Ethernet payload is 46 bytes. If the encapsulated packet is smaller than 46 bytes, **padding bytes** are added to the payload so the Ethernet frame reaches the minimum size of 64 bytes.

MAC Addresses:

MAC Addresses are 6 byte (48 bit) physical addresses assigned to a device when it is made (written in hexadecimal). They are globally unique - the first 3 bytes are the **OUI (Organizationally Unique Identifier)** which identifies the vendor that produced the device, the last 3 bytes are unique to the device itself.

Hexadecimal uses 16 possible digits:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

A switch learns MAC addresses **dynamically** by examining the source MAC address of frames it receives and associating that MAC address with the interface the frame entered on.

On Cisco switches, dynamically learned MAC address entries are removed after 5 minutes (300 seconds) of inactivity by default. This process is known

as MAC address **aging**.

To view the MAC address table on Cisco IOS, run:

- show mac address-table

To manually clear all dynamic MAC address entries, run:

- clear mac address-table dynamic

To clear a specific dynamic MAC address entry based on its MAC address:

- clear mac address-table dynamic address *MAC-address*

To clear dynamic MAC address entries associated with a specific interface:

- clear mac address-table dynamic interface *interface*

Example:

- PC1 sends a unicast frame to PC2 (Destination: .0002, Source: .0001).
- When SW1 receives the frame on F0/1, it examines the source MAC address and adds it to its MAC address table, mapping .0001 to F0/1. Because the switch learned this entry automatically from received traffic, it is called a **dynamically learned MAC address**.
- Assuming these devices are newly connected and SW1 has not yet learned PC2's MAC address, SW1 checks its MAC address table for destination .0002 and finds no matching entry. This makes the frame an **unknown unicast frame**.
- Because SW1 does not know which interface leads to .0002, it floods the frame out all appropriate interfaces except the interface it was received on. In this example, the frame is sent out F0/2 and F0/3. PC3 receives the frame but drops it because the destination MAC address does not match its own. PC2 receives the frame and processes it.
- If PC2 replies to PC1 with a frame (Destination: .0001, Source: .0002), SW1 receives the frame on F0/2. It first examines the source MAC address and adds .0002 = F0/2 to its MAC address table.
- SW1 then checks the destination MAC address .0001. Because it previously learned that .0001 is reachable through F0/1, SW1 forwards the frame only out F0/1 instead of flooding it. This is referred to as a **known unicast frame**.

ARP:

ARP (Address Resolution Protocol) is used to discover the Layer 2 address (MAC address) associated with a known Layer 3 address (IPv4 address).

ARP primarily uses two messages:

- ARP Request: Sent as a broadcast to ask which device owns a specific IPv4 address.
- ARP Reply: Sent as a unicast back to the requesting device and contains the requested MAC address.

The information learned through ARP is stored in the device's ARP table.

To view the ARP table on Windows, run:

- arp -a

On modern Linux systems, the commonly used command is:

- ip neigh



```
Interface: 10.0.2.15 --- 0x4
Internet Address      Physical Address      Type
10.0.2.2              52-54-00-12-35-00    dynamic
10.0.2.255            ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-10    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static

C:\Windows\system32>
```

In ARP table output:

- Internet Address = IPv4 address (Layer 3)
- Physical Address = MAC address (Layer 2)
- Dynamic = Entry learned through ARP
- Static = Entry that was not dynamically learned through ARP

To view the ARP table on Cisco IOS, run:

- show arp

Ping:

Ping is a network utility used to test IP reachability between devices and measure round-trip time.

Ping uses ICMP (Internet Control Message Protocol) and primarily involves two messages:

- ICMP Echo Request: Sent by the device initiating the ping.
- ICMP Echo Reply: Sent back by the destination if it receives the request and is able to respond.

To ping another device, run:

- ping *IP-address*

IPv4:

The Network Layer (Layer 3) provides connectivity between end hosts on different networks (between LANs). It provides logical addressing (IP addresses), and path selection between source and destination.

Decimal is base 10:

1s, 10s, 100s, 1000s, etc.

Binary is base 2:

1, 2, 4, 8, 16, 32, etc.

An IPv4 address is comprised of 32 bits, separated by each 8 bit set - referred to as an octet.

Example: xxx.xxx.xxx.xxx

(Each xxx is 8 bits - an octet - which can be represented by binary or decimal.)

Binary to Decimal examples:

1111 = 15

1011 = 11

0101 = 5

Decimal to Binary examples:

29 = 11101

72 = 1001000
195 = 11000011

IPv4 Classes

Class A	0.0.0.0 - 127.0.0.0	Used for large businesses.
Class B	128.0.0.0 - 191.0.0.0	Used for medium-sized businesses.
Class C	192.0.0.0 - 223.0.0.0	Used for small businesses and home networks.
Class D	224.0.0.0 - 239.0.0.0	Used for Multicast addresses.
Class E	240.0.0.0 - 255.0.0.0	Used for Experimental purposes.

Net Masks:

A subnet mask (netmask) tells you which part of an IPv4 address identifies the network and which part identifies the host.

Example:

IP address 192.168.1.10

Subnet mask 255.255.255.0

192.168.1 = network portion

10 = host portion

CIDR Notations:

A CIDR notation is just a shorter way of writing the same subnet mask.

Example:

255.255.255.0 = /24

The /24 means the first 24 bits of the 32-bit IPv4 address are network bits.

Remember:

Netmask = dotted-decimal representation

CIDR = number of network bits

They are two ways of expressing essentially the same information.

Network & Host Addresses:

A **network address** identifies the network itself.

Example:

192.168.1.0/24

Network address = 192.168.1.0

It represents the entire 192.168.1.0/24 network and is not normally assigned to a host.

A **host address** identifies a specific device inside that network.

Examples:

192.168.1.1

192.168.1.25

192.168.1.200

These are host addresses within the 192.168.1.0/24 network.

- **Network address = identifies the network**
- **Host address = identifies a device within that network**

Calculating network and broadcast addresses:

192.168.1.0/24

Network address = 192.168.1.0

Identifies the network itself.

Broadcast address = 192.168.1.255

Used to send traffic to all hosts on that network.

Usable host range = 192.168.1.1 - 192.168.1.254

The network and broadcast addresses cannot normally be assigned to hosts.

Calculating the number of usable hosts based on an address:

To calculate the number of usable host addresses, use:

- **2^h-2**
- (h = host bits)

So for a /24 network, 24 bits are used to identify the network, leaving 8 host bits, so 2⁸-2= 254 usable host addresses.

Exploring IP addressing in the Cisco CLI:

To confirm status and IP address of device interfaces, run:

- show ip interface brief

```
SAT01-SW01(config)#do show ip int br
Interface                IP-Address      OK? Method Status
Protocol
GigabitEthernet1/0/1    unassigned      YES unset  up
GigabitEthernet1/0/2    unassigned      YES unset  up
GigabitEthernet1/0/3    unassigned      YES unset  up
GigabitEthernet1/0/4    unassigned      YES unset  down
GigabitEthernet1/0/5    unassigned      YES unset  down
GigabitEthernet1/0/6    unassigned      YES unset  down
GigabitEthernet1/0/7    unassigned      YES unset  down
GigabitEthernet1/0/8    unassigned      YES unset  down
GigabitEthernet1/0/9    unassigned      YES unset  down
GigabitEthernet1/0/10   unassigned      YES unset  down
GigabitEthernet1/0/11   unassigned      YES unset  down
GigabitEthernet1/0/12   unassigned      YES unset  down
GigabitEthernet1/0/13   unassigned      YES unset  down
GigabitEthernet1/0/14   unassigned      YES unset  down
GigabitEthernet1/0/15   unassigned      YES unset  down
GigabitEthernet1/0/16   unassigned      YES unset  down
GigabitEthernet1/0/17   unassigned      YES unset  down
GigabitEthernet1/0/18   unassigned      YES unset  down
GigabitEthernet1/0/19   unassigned      YES unset  down
GigabitEthernet1/0/20   unassigned      YES unset  down
GigabitEthernet1/0/21   unassigned      YES unset  down
--More--
```

The output of show ip interface brief provides 6 columns of information:

Interface	Lists the interfaces on the device.
IP-Address	Displays the IP address assigned to each interface.
OK?	Legacy field; generally not important for modern troubleshooting.
Method	Indicates how the IP address was assigned, such as manually or through DHCP.
Status	Indicates the Layer 1 status of the interface.
Protocol	Indicates the Layer 2 status of the interface.

Remember: Administratively down means an interface has been disabled with the shutdown command. Cisco router interfaces are administratively down by default. In contrast, Cisco switch interfaces are NOT administratively down by default.

To configure a specific interface, run:

- interface *interface-id*

If done successfully, (config-if) will be displayed confirming you’re in interface configuration mode.

To set the IP address for an interface:

- ip address *ip-address subnet mask*

To enable the interface:

- no shutdown

To verify your configurations have taken effect, run:

- show ip interface brief

To view interface information:

- show interfaces *interface-id*

With this output, you can view the layer 1/2 status, hardware speed information, MAC address of the interface, IP address of the interface, and more.

To view interface description:

- show interfaces description

To configure interface description:

- description *description*

There aren't any rules governing how the description should be formatted, but encasing the description in something like hashtags can serve to make it neater/more noticeable (e.g., ## to SW1 ##).

Exploring switch interfaces in the Cisco CLI:

Switch interfaces are typically in an up/up state by default (if there is something plugged into that interface). Switch interfaces that are not connected to another device will display down/down.

To view port/name/status/vlan/duplex/speed/type, run:

- show interfaces status

Duplex is auto on Cisco switches by default, meaning it will negotiate with the neighboring device regarding whether they will use full or half.

To manually set the speed on an interface:

- speed *10-100-auto-etc*

To manually set the duplex on an interface:

- duplex *auto-full-half*

To configure multiple interfaces at once:

- interface range *f0/# - #*

To select specific interfaces to configure:

- interface range *f0/# - #, f0/# - #*

To shutdown unused interfaces:

- shutdown

To view interface errors:

- show interfaces

Half duplex: The device cannot send and receive data at the same time. If it is receiving a frame, it must wait before sending a frame. Devices attached to a hub must operate on half duplex.

Full duplex: The device can send and receive data at the same time. It does not have to wait.

Collisions occur in a half-duplex environment, when more than one frame is traveling on any one medium. A collision causes the frames to either be delivered fragmented/broken or unable to be delivered altogether.

To avoid such collisions, Ethernet devices have something called **CSMA/CD (Carrier Sense Multiple Access with Collision Detection)**.

How CSMA/CD works:

- Before sending frames, devices ‘listen’ to the collision domain until they detect that other devices are not sending.
- If a collision does occur, the device sends a jamming signal to inform the other devices that a collision happened.
- Each device will wait a random period of time before sending frames again.
- The process repeats.

Speed/Duplex Auto Negotiation:

Interfaces that can run at different speeds (10/100, or 10/100/1000) have default settings of speed auto and duplex auto.

Interfaces ‘advertise’ their capabilities to the neighboring device, and they negotiate the best speed and duplex settings they are both capable of.

What if auto negotiation is disabled on the device connected to the switch?

The switch will try to sense the speed that the other device is operating at. If it fails to sense the speed, it will use the slowest supported speed on its interface (10 Mbps on a 10/100/1000 interface).

If the speed is 10 or 100 Mbps, the switch will use half duplex - - if the speed is 1000 Mbps or greater, the switch will use full duplex.

A duplex mismatch will cause collisions to occur.

```
SAT01-SW01(config)#do show interfaces
GigabitEthernet1/0/1 is up, line protocol is up (connected)
  Hardware is Lance, address is 0004.9ac8.4201 (bia 0004.9ac8.4201)
  MTU 1500 bytes, BW 1000000 Kbit, DLY 10 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 04:00:00
  last input 00:00:08, output 00:00:05, output hang never
  last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    956 packets input, 193351 bytes, 0 no buffer
      Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 0 multicast, 0 pause input
    0 input packets with dribble condition detected
  2357 packets output, 263570 bytes, 0 underruns
--More--
```

Understanding Interface Errors:

Runts	Frames smaller than the minimum Ethernet frame size of 64 bytes.
Giants	Frames larger than the standard maximum Ethernet frame size of 1518 bytes.
CRC	Frames that failed the CRC check performed using the FCS trailer, usually indicating that the frame was corrupted during transmission.
Frame	Frames that have a framing error, meaning the received frame does not conform to the expected Ethernet frame structure.
Input errors	A cumulative count of various errors that occurred while receiving frames, including errors such as runts, giants, CRC errors, and frame errors.
Output errors	A cumulative count of errors that prevented frames from being successfully transmitted out of the interface.

IPv4 Header:

Version

Length	4 bits
Function	Identifies the version of IP used.
Format	IPv4 = 4 (0100) IPv6 = 6 (0110)

Internet Header Length (IHL)

Length	4 bits
Function	Indicates the total length of the IPv4 header. Necessary because the Options field varies in length.
Format	5 - 15

The Format value represents the header length in 4-byte (32-bit) increments.

Example: A value of 5 means $5 \times 4 = 20$ bytes.

- Minimum value: 5 = 20-byte header
- Minimum IPv4 header length: 20 bytes
- Maximum value: 15 = 60-byte header
- Maximum IPv4 header length: 60 bytes

Differentiated Services Code Point (DSCP)

Length	6 bits
Function	Used for QoS (Quality of Service). Allows traffic to be given different levels of priority.
Format	6 bit value ranging from 0-63

Explicit Congestion Notification (ECN)

Length	2 bits
Function	Optional feature that requires both endpoints, as well as the underlying network infrastructure, to support it. Provides end to end notification of network congestion without dropping packets.
Format	2-bit value used to indicate whether ECN is supported and whether congestion has been experienced.

Total Length

Length	16 bits
Function	Indicates the total length of the IPv4 packet, including the IPv4 header and its payload.
Format	16-bit value; the value represents the packet length in bytes.

- Minimum value of 20 bytes (IPv4 header with no encapsulated data).
- Maximum value of 65535 (maximum 16-bit value).

Identification

Length	16 bits
Function	Used when an IPv4 packet is fragmented - fragments belonging to the same original packet use the same Identification value so the receiving host can determine which fragments belong together.
Format	16-bit identification value, fragments of the same original packet carry the same identification value.

Fragmentation can occur when a packet is larger than the **MTU (Maximum Transmission Unit)** of a link and fragmentation is permitted. A common

Ethernet MTU is **1500 bytes**. Fragments are reassembled by the destination host.

Flags

Length	3 bits
Function	Used to control and identify IPv4 fragmentation.
Format	Bit 0: Reserved and set to 0. Bit 1: Don't Fragment (DF) indicates that the packet must not be fragmented when set to 1. Bit 2: More Fragments (MF) set to 1 when additional fragments follow and set to 0 for the final fragment. An unfragmented packet has MF set to 0.

Fragment Offset

Length	13 bits
Function	Allows the destination host to correctly reassemble fragments even if they arrive out of order, because this field indicates where a fragment belongs within the original unfragmented packet.
Format	Varies

Time to Live

Length	8 bits
Function	Prevents packets from circulating indefinitely due to routing loops. In practice, TTL functions as a hop limit. Each router that forwards the packet decreases the TTL by 1. If the TTL reaches 0, the router drops the packet.
Format	8-bit value ranging from 0-255.

64 is the recommended TTL default however, defaults vary between operating systems and devices.

Protocol

Length	8 bits
Function	Identifies the protocol encapsulated within the IPv4 packet so the receiving host knows how to process the payload.
Format	Each protocol is represented by an assigned numeric value.

Value of 1: ICMP

Value of 6: TCP

Value of 17: UDP

Value of 89: OSPF

Header Checksum

Length	16 bits
Function	A calculated checksum used to check for errors in the IPv4 header.
Format	Varies

When a router receives a packet, it calculates the checksum of the header and compares it to the one in this field of the header. If they do not match, the router drops the packet.

Important: Both TCP and UDP have their own checksum fields to detect errors in the encapsulated data. IP relies on the encapsulated protocol to detect errors in the encapsulated data.

Source/Destination IP address

Length	32 bits
Function	Identifies the IPv4 address of the packet's sender/intended destination.
Format	32-bit IPv4 address

Options

Length	0-320 bits
Function	Provides optional IPv4 features that are rarely used in normal modern traffic.
Format	Varies

Important: If the IHL is greater than 5, it means that Options are present.